ECAP509: SOFTWARE ENGINEERING

UNIT-1

1. **Which one is system software?**
   a. MS office
   b. Windows XP
   c. Chrome
   d. all of above

2. **What are the software crises?**
   a. Quality
   b. Size
   c. Cost
   d. All of above

3. **Which one is application software ________**
   a. Sound drivers
   b. Operating system
   c. LAN drivers
   d. Internet explorer

4. **Software is ________**
   a. Documentation and configuration of data
   b. Set of programs
   c. Set of programs, documentation & configuration of data
   d. None of the mentioned

5. **Characteristics of a software is ___________**
   a. Operational
   b. Transitional
   c. Maintenance
   d. All of above

6. **Software processes resources are ________**
   a. SPICE
   b. CMM
   c. PSP
   d. all of above

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7. **Software process activities include**
   a. Quality
   b. Size
   c. Cost
   d. All of above

8. **The process of developing a software product using software engineering principles and methods is referred to as _______**
   a. Software Engineering
   b. Software Evolution
   c. System Models
   d. Software Models

9. **SDLC stands for**
   a. System Development Life cycle
   b. Software Design Life Cycle
   c. Software Development Life Cycle
   d. System Design Life Cycle

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UNIT-2

1. **Which is not a SDLC phase?**
a. Design
b. Coding
c. Testing
d. Coupling

2. **Which is not a part of feasibility study?**
a. Time
b. Operational
c. **Module**
d. Cost

3. **Which is not a testing type?**
a. Unit
b. Beta
c. Cohesion
d. Black box

4. **Prototype model is ____**
a. Suitable where the project's requirements are not known in detail to customer
b. It is a demo implementation of the actual product or system
c. Used to allow the users evaluate developer proposals
d. All of above

5. **Which is not a part of prototype model phases?**
a. Requirement
b. Design
c. Process
d. Implementation

6. **Which is not a prototype model type?**
a. Evolutionary
b. Beta
c. Incremental
d. Extreme

7. **Waterfall model is ____**
a. It is linear sequential model
b. It is the classic lifecycle model
c. Introduced by Winston W. Royce in 1970
d. All of above

8. Stages in Waterfall model are _____

a. Design
b. Implementation
c. Testing and Deployment
d. All of above

9. Information is gathered in ________ phase

a. Design
b. Coding
c. Requirement analysis
d. Maintenance

10. Incremental Model is ____

a. Broken down into multiple standalone modules of software development cycle.
b. Multiple development cycles take place
c. Each iteration passes through the requirements, design, coding and testing phases
d. All of above

UNIT-3

1. **Software requirement gathering from client, analyze and document is known as**
   a. Requirement Gathering
   b. Requirement Engineering
   c. Feasibility Study
   d. System Requirements Specification

2. **The objective of requirement engineering is to develop and maintain_____ document.**
   a. Feasibility Study
   b. System Requirements Specification
   c. Requirement Gathering
   d. Software Requirement Validation

3. **Which one is a correct requirement type?**
   a. Reliability
   b. Availability
   c. Usability
   d. All of the above

4. **Which is not step of requirement engineering?**
   a. Elicitation
   b. Documentation
   c. Design
   d. Analysis

5. **What are the activities involve in requirement gathering?**
   a. Interview
   b. Survey
   c. Group discussion
   d. All of above

6. **Product requirement include _____**
   a. Efficiency
   b. Operational
   c. Development
   d. All of above

7. **What are the types of requirements?**
   a. Functional
   b. Non–functional
   c. System
   d. All of above

8. Requirement Engineering process activities are_

a. Feasibility Study
b. Requirement Elicitation and Analysis
c. Software Requirement Validation
d. All of above

9. Economic Feasibility deals with.

a. Technical activity
b. Operational activity
c. Cost activity
d. None of above

10. Requirement Elicitation activities are ____

a. Organizing Requirements
b. Negotiation & discussion
c. Documentation
d. All of the above

11. Which is not Requirement Elicitation Techniques?

a. Surveys
b. Questionnaires
c. Design
d. Task analysis

12. Requirements Validation Techniques are ____

a. Requirements reviews/inspections
b. Prototyping
c. Test-case generation
d. All of above

13. How the interviews held between two persons across the table is ___

a. Non-structured
b. One-to-one
c. Written
d. Group

14. Which is not a feasibility study?

a. Operational
b. Finance
c. Modular
d. Social

UNIT-4

1. Functional and non-functional requirement are the part of ____

a. Design phase
b. SRS
c. Testing
d. Maintenance

2. Functional requirements are _____

a. Defines a system or its component
b. Describe a particular behavior of function of the system when certain conditions are met
c. "What the System Should Do"
d. All of above

3. Functional requirements not include _______

a. Business Rules
b. Administrative functions
c. Reusability
d. Authentication

4. Audit Tracking, External Interfaces and historical data is part of ____

a. Nonfunctional requirement
b. Functional requirement
c. All of the above

5. Nonfunctional requirements are ______

a. Defines the quality attribute of a software system
b. Represent a set of standards used to judge the specific operation of a system
c. Essential to ensure the usability and effectiveness of the entire software system
d. All of above

6. Non Functional Requirement include

a. Scalability
b. Performance
c. Reusability
d. All of above

7. Each request should be processed within predefined time is _______

a. **Nonfunctional requirement**
b. Functional requirement
c. All of the above

8. SRS stands for ____

a. System Requirements Specification
b. **Software Requirements Specification**
c. Software Requirements support
d. None of above

9. Software Requirements Specification is _____

a. Developed with its functional and non-functional requirements
b. Must address the needs and expectations of the users
c. It serves a different purpose and mentions the contrast between the customer and the developer.
d. **All of above**

10. Characteristics of a SRS _______

a. Correct
b. Unambiguous
c. Complete
d. **All of above**

11. As per IEEE which is not a part of Structure of Document

a. Purpose
b. Scope
c. **Testing**
d. Product Functions

12. The SRS document is also known as _____________ specification

a. **Black-box**
b. White-box
c. Grey-box
d. None of above

13. Which of the following is not included in SRS?

a. Functionality
b. Performance
c. Design solutions
d. External Interfaces

14. What is the first step of requirement elicitation?

a. Identifying Stakeholder
b. Listing out Requirements
c. Requirements Gathering
d. All of above

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UNIT-5

1. Software design process is ____

a. Converts the ''what'' of the requirements to the ''how'' of the design
b. The first step in SDLC (Software Design Life Cycle)
c. Process of problem solving and planning for a software solution
d. All of above

2. Which is not Software Design Level?

a. Interface Design
b. Architectural Design
c. Use case design
d. High-level Design

3. Software Design Concept include ____

a. Modularity
b. Refinement
c. Abstraction
d. All of above

4. How to measure quality and interaction between modules

a. Coupling
b. Cohesion
c. Both coupling and cohesion
d. None of above

5. Which is not a type of coupling?

a. Content
b. Common
c. Logical
d. Control

6. Which is not a type of cohesion?

a. Temporal
b. Procedural
c. Communicational
d. Stamp

7. What are the types of Data Flow Diagram?

a. Physical
b. Logical
c. ***Both physical and logical***
d. None of above

8. Which is not a part of DFD symbol?

a. Processes
b. Data stores
c. ***Decision table***
d. External entities

9. Use case and activity diagram is part of _____

a. Structure diagram
b. ***Behavior diagram***
c. Both Behavior diagram and Structure diagram
d. None of above

10. Which symbol is not a part of use case diagram?

a. Actor
b. ***Decision making***
c. System
d. Use case

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UNIT-6

1. Which is part of user interface?

a. Command Line Interface
b. Graphical User Interface
c. Both Command Line Interface and Graphical User Interface
d. None of above

2. Which is not a part of Graphical User Interface?

a. Icon
b. Button
c. Command
d. Cursor

3. Command Line Interface is ___

a. Minimal memory usage
b. Great for slow-running computers
c. This method relies primarily on the keyboard
d. All of above

4. Which is not a part of Command Line Interface?

a. UNIX
b. DOS
c. LINUX
d. None of above

5. Which is not a part of Graphical User Interface?

a. Windows XP
b. UNIX
c. LINUX
d. None of above

6. Graphical User Interface is ____

a. Memorizing command lists is not necessary
b. Allows for running multiple applications, programs, and tasks simultaneously
c. Follow WYSIWYG (What You See Is What You Get)
d. All of above

7. Which is not a part of Interface Design Model?

a. User model
b. Design model
c. SDLC model
d. Mental model

8. How many phases in Interface design process?

a. 2
b. 3
c. 4
d. 5

9. Which of the following devices are mainly responsible for the user interface?

a. Input and output devices
b. Memory devices
c. Processor
d. None of the above

10. Which is not a part of Interface design process?

a. Interface design
b. Testing
c. Interface construction
d. User, task and environment analysis

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UNIT-7

1. What are the common types of error in coding?

a. Memory leaks
b. Null dereferencing
c. Null dereferencing
d. All of above

2. Which is not a syntax error?

a. printf (“this is code’);
b. printf (“this is code”);
c. printf (‘this is code’);
d. none of above

3. Which is not a coding error?

a. Logical error
b. Lack of unique addresses
c. Transformation
d. None of the above

4. A good coding practice is?

a. Portability
b. Generality
c. Enhance Code Readability
d. All of above

5. Coding is ___

a. To translate the design of system into a computer language format:
b. To reduce the cost of later phases
c. Making the program more readable
d. All of above

6. Coding standards are __

a. Help to ensure safety, security, and reliability.
b. Collections of coding rules, guidelines, and best practices
c. Compliance with industry standards
d. All of above

7. Which is not a part of coding standard?

a. **ISO 2010**
b. IEC 61508
c. EN 50128
d. IEC 62061

8. Which coding standard is for Railway applications?

a. ISO 26262
b. IEC 62061
c. ***EN 50128***
d. None of the above

9. Lockheed Martin developed ____

a. MISRA
b. ***JSFC++***
c. AUTOSAR
d. All of above

10. MISRA coding standard is used for __

a. Fighter Air Vehicle
b. ***Automotive embedded software industry***
c. Toy industry
d. None of above

11. What are the different type of documents?

a. Operational manual
b. List of Known Bugs
c. User manual
a. **All of above**

12. Code can be reuse in case _

a. Easily extended and adapted for the new application.
b. Ported to different hardware if needed.
c. Shown to be free from defects
d. ***All of above***

13. Which is not a Code reusability methods?

b. Inheritance
c. Functions
d. ***Classes***
e. Forking

14. What are different Code reusability methods?

a. Functions
b. Libraries
c. Forking
d. All of above

15. Function call is part of __

a. Testing
b. Classes
c. Code reusability
d. None of above

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UNIT-8

1. Good test design supports ____

a. Defining and improving quality related processes and procedures
b. Evaluating the quality of the product with regards to customer expectations and needs
c. Finding defects in the product
d. All of above

2. What are the Prerequisites of test design?

a. Appropriate specification
b. Risk and complexity analysis.
c. Historical data of your previous developments
d. All of above

3. Manual Testing includes.

a. Unit testing
b. Smoke testing
c. Black box testing
d. None of above

4. What are different Types of Software Testing?

a. Functional Testing
b. Non-Functional Testing
c. Maintenance
d. All of above

5. Which is not a part of Functional testing?

a. Localization
b. Globalization
c. Black box testing
d. None of above

6. What are the Types of Test Plan?

a. Master Test Plan
b. Phase Test Plan
c. Testing Type Specific Test Plans
d. All of above

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7. As per IEEE 829 standard which is not part of test plan?

a. Design the Test Strategy
b. Feasibility study
c. Define the Test Objectives
d. Define Test Criteria

8. What are the best practice for writing Test Case?

a. Create Test Case with End User in Mind
b. Avoid test case repetition
c. Do not assume
d. All of above

9. Equivalence Partition is part of ____

a. Design
b. Testing Technique
c. Maintenance
d. All of above

10. What are the types of test cases?

a. User Interface Test Cases
b. Performance Test Cases
c. Integration Test Cases
d. All of above

11. Which is not part of test case?

a. Usability
b. Database
c. Design
d. Security

12. Which is test case template fields?

a. Test Priority
b. Name of the Module
c. Test Designed by
d. All of above

13. Status (Fail/Pass) is part of __

a. Design
b. Test case template fields
c. Maintenance

d. None of above

14. Test case is _____

a. Design activity
b. Feasibility study
c. Contains test steps, test data, pre-condition, post condition
d. None of above

15. Why test case required?

a. To require consistency in the test case execution
b. To make sure a better test coverage
c. To avoid training for every new test engineer on the product
d. All of above

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UNIT-9

1. Black box testing is ____

a. Focuses on the functional requirements of the software
b. No knowledge required for internal code structure, implementation details and internal paths
c. It is used to test the system against external factors responsible for software failures
d. All of above

2. What are the types of Black Box Testing?

a. Functional testing
b. Non-functional testing
c. Regression testing
d. All of above

3. Boundary Value Analysis is part of _

a. White box testing
b. Black box testing
c. Unit testing
d. None of above

4. Which is not part of Black Box Testing Techniques?

a. Cause effect Graphing
b. Compatibility testing
c. Load testing
d. Syntax Driven Testing

5. Generating test cases and Identification of equivalence class are part of __

a. Cause effect Graphing
b. Equivalence Class Partitioning
c. Syntax Driven Testing
d. None of above

6. Cause effect graphing is __

a. Identify inputs (causes) and outputs (effect).
b. Develop cause effect graph.
c. Transform the graph into decision table
d. All of above

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7. yntax Driven Testing and State Transition Testing is part of ____

a. Integration testing
b. Unit testing
c. White box testing
d. Black box testing

8. What are different Loop Testing __?

a. Simple
b. Nested
c. Concatenated
d. All of above

9. NUnit and PyUnit is part of ____

a. Integration testing
b. White box testing
c. Unit testing
d. Black box testing

10. What are the types of Smoke testing?

a. Manual method
b. Automation method
c. Hybrid method
d. All of above

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UNIT-10

1. Unit testing is _______

a. It is a White Box testing technique
b. It involves the testing of each unit
c. It is done during the development of an application by the developers
d. All of above

2. What are the steps for work flow of unit testing?

a. Creating test cases
b. Reviewing test cases
c. Base lining test cases
d. All of above

3. Types of Unit testing are _____

a. Manual
b. Automated
c. Both manual and automated
d. None of above

4. White box testing and Grey box testing is part of __

a. Stress testing
b. Regression testing
c. Unit testing
d. None of above

5. Tools used in Unit Testing are __________

a. PHPUnit
b. EMMA
c. JMockit
d. All of above

6. Incremental Testing and Stubs and Drivers are part of ___

a. Unit testing
b. White box testing
c. Integration testing approaches
d. None of above

7. What are the advantages of Top-Down Integration Testing?

a. Few or no drivers needed.
b. Fault Localization is easier.
c. Possibility to obtain an early prototype
d. All of above

8. Which is not part of Integration testing approaches _____

a. Top-Down Integration Testing
b. Black box testing
c. Mixed Integration Testing
d. Incremental Testing

9. System testing is _______

a. Falls under the black box testing
b. It is to evaluate the end-to-end system specifications
c. Performed to validates the complete and fully integrated software product
d. All of above

10. Recovery testing and Migration testing are used in ____

a. Unit testing
b. Stress testing
c. System testing
d. All of above

11. Alpha and beta testing is used in ___

a. Manual
b. Automated
c. User Acceptance testing
d. None of above

12. Tools used in User Acceptance testing are __

a. Watir
b. Fitness
c. Both watir and fitness
d. None of above

13. Regression testing is _______

a. Re-execute or re-running functional and non-functional tests
b. To make sure that new code changes should not have side effects on the existing functionalities
c. Changes in Existing functionality
d. All of above

14. What are the Steps for regression testing?

a. Detect Changes in the Source Code
b. Prioritize Those Changes and Product Requirements
c. Determine Entry Point and Entry Criteria
d. All of above

15. Partial Testing and Unit Testing is part of ___

a. Stress testing.
b. Load testing.
c. Regression Testing
d. All of above

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UNIT-11

1. What are the reasons for bug?

a. Wrong coding
b. Missing coding
c. Extra coding
d. All of above

2. What are the classifications of software bugs?

a. Software defects by priority
b. Software defects by severity
c. Software defects by nature
d. All of above

3. Performance defects and Usability defects are part of ___

a. Software defects by severity
b. Software defects by nature
c. Software defects by priority
d. None of above

4. Software defects by priority consist of __

a. Critical defects
b. High-severity defects
c. Medium-severity defects
d. None of above

5. Software defects by priority are based upon _____

a. Severity
b. Priority
c. By nature
d. All of above

6. Which is not part of Bug life cycle states?

a. Open
b. Fixed
c. Design
d. Retest

7. Assign, fixed and retest is part of ___

a. Software defects by priority
b. ***Bug life cycle***
c. Software defects by nature
d. All of above

8. Which is not bug tracking tool?

a. Zoho tracker
b. ***Bootstrap***
c. Monday
d. JIRA

9. Trac is developed in _

a. Java
b. C++
c. ***Python***
d. None of above

10. eTraxis support ___

a. MySQL
b. PostgreSQL
c. MS Server
d. ***All of above***

11. Bug tracking system is __

a. Report of specific bug file
b. It is process of logging and monitoring bugs
c. Option to change the status of the bug
d. ***All of above***

12. Which is not Bug life cycle state?

a. Open
b. ***Feasibility study***
c. Reset
d. None of above

13. eTraxis, Bugnet and Axosoft are __

a. Design tool
b. Testing tools
c. ***Bug tracking tool***

d. None of above

14. Mantis is ___

a. Open source tool
b. Bug tracking tool
c. Works with multiple databases like MySQL, PostgreSQL
d. All of above

15. Gantt chart and calendar are used in _

a. Trac
b. RedMine
c. JIRA
d. All of above

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UNIT-12

1. Software maintenance is used for __

a. Removal of Outdated Functions
b. Performance Improvement
c. Market Conditions
d. Above all

2. Which is not type of Software Maintenance?

a. Unit maintenance
b. Adaptive maintenance
c. Perfective maintenance
d. Above all

3. Software Maintenance Activities includes _

a. Design
b. Implementation
c. System testing
d. Above all

4. Software Support encompass __

a. Operation
b. Logistics Management
c. Modification
d. Above all

5. Software supportability process not include __

a. Program planning and control
b. System testing
c. Preparation and evaluation of alternatives
d. Determination of SAS requirements

6. What are the Software Supportability Principles

a. Check the "ilities"
b. Manage Change
c. Control Quality
d. Above all

7. What are the Steps of business process reengineering?

a. Analyze them and find any process gaps or disconnects
b. Look for improvement opportunities and validate them
c. Design a cutting-edge future-state process map
d. Above all

8. Which is not part of Reengineering project guidelines __

a. Designing solution
b. Developing business case justification
c. Redesign the system
d. Developing solution

9. What are the Motivations for reengineering?

a. Reduce external competition pressure
b. Reverse erosion of market share
c. Respond to emerging market opportunities
d. Above all

10. What are the parameters for Economics of Re-Engineering?

a. Current annual maintenance cost for an application.
b. Current annual operation cost for an application.
c. Estimated reengineering costs
d. Above all

UNIT-13

1. What are the characteristics of software Metrics?

a. Quantitative
b. Applicability
c. Repeatable
d. Above all

2. Which is not type of Software Metrics?

a. Internal metrics
b. Product metrics
c. External metrics
d. Project metrics

3. Product metrics classes are _

a. Dynamic
b. Static
c. Both dynamic and static
d. None of above

4. Software Product Metrics include

a. Fan-in/Fan-out
b. Length of code
c. Cyclomatic complexity
d. All of above

5. Which is not part of Product Quality Metrics?

a. Defect Density
b. Product metrics
c. Customer Problems
d. Customer Satisfaction

6. What are the point scale for Customer Satisfaction?

a. Very satisfied
b. Satisfied
c. Neutral
d. Above all

7. Why we need of software measurement?

a. Create the quality of the current product or process.
b. Anticipate future qualities of the product or process.
c. Enhance the quality of a product or process.
d. Above all

8. Software Measurement include?

a. Direct Measurement
b. Indirect Measurement
c. Both direct and indirect measurement
d. None of above

9. Which is not activities of a measurement process

a. Testing
b. Formulation
c. Collection
d. Feedback

10. How many types of data functions

a. 4
b. 3
c. 2
d. 5

11. Transaction Functions include?

a. External Inputs
b. External Outputs
c. External Inquiries
d. All of above

12. Which is not types of functional point attributes?

a. EQ
b. AI
c. EI
d. ILF

13. What are the shortcomings of function point metric?

a. Conversion need
b. Low Accuracy
c. Time consuming
d. All of above

14. What are the parameter for computing the unadjusted function point (UFP)

a. Number of inputs
b. Number of inquiries
c. Number of files
d. All of above

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UNIT-14

1. What are the steps for Software process improvement?

a. Current Situation Evaluation
b. Improvement Planning
c. Improvement Implementation
d. Above all

2. Software process improvement methods are __

a. Inductive
b. Prescriptive
c. Both inductive and prescriptive
d. None of above

3. What are the advantages of Software process improvement?

a. Cost reduction
b. Competitive Edge
c. Standardization and Process consistency
d. All of above

4. What are the Demotivation factors for Software process improvement?

a. Budget Constraints
b. Time pressure
c. Inadequate metrics
d. All of above

5. CMMI and SPICE are part of __

a. Prescriptive
b. Inductive
c. Both inductive and prescriptive
d. None of above

6. Which is not Software Process Improvement Model?

a. SIX SIGMA
b. IBM 02
c. SPICE
d. TickIT Guide ISO 9001

7.What are the areas of interest for CMMI?

a. Product and service development
b. Service establishment, management, and delivery
c. Product and service acquisition
d. ***All of above***

8. BOOTSTRAP is __

a. Asian method
b. ***European method***
c. American method
d. All of above

9. What are the principal area of CMM?

a. Organization and resource management
b. Software engineering process and its management
c. Tools and technology
d. ***All of above***

10. Which is not part of CMM Maturity Level?

a. Repeatable
b. Defined
c. ***Testing***
d. Initial Managed

11. CMMI V2.0 Key Improvements are __

a. Improve Business Performance
b. Build Agile Resiliency and Scale
c. Build Agile Resiliency and Scale
d. ***Above all***

12. Which is not types of CMMI Appraisals?

a. Benchmark Appraisal
b. ***System design appraisal***
c. Sustainment Appraisal
d. Action Plan Reappraisal

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13. What are the CMMI Maturity Levels?

a. Repeatable
b. Defined
c. Initial Managed
d. ***All of above***

14. CMMI model components are __

a. Process areas
b. Goals
c. Practices
d. ***All of above***

15. Process Management and Project Management are part of __

a. Practices
b. Goals
c. ***CMMI Process Areas***
d. None of above

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